

Olympiad Test Paper — Social Science (Class 7) — Paper 4

Chapter 3: Climates of India

Paper 4 — (progressively harder)

Subject	Social Science (NCERT Class 7)
Chapter	3 — Climates of India
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: confusing weather with climate, blaming equatorial heat on nearness to the Sun, ignoring altitude and the moderating sea, reversing which monsoon is wet, mistaking the calm eye for the safe part of a cyclone, and assuming the wet windward and dry leeward sides of a range rain equally. Do not write on this paper — mark answers on your answer sheet.

1. Three reports describe a single hill town: (i) 'It snowed here this morning,' (ii) 'Winters here are reliably cold and snowy every year,' (iii) 'The thermometer reads 2 degrees right now.' Which report alone describes climate rather than weather?

- (A) Only (i), because snow is always a sign of a place's climate
- (B) Only (iii), because a thermometer reading is the most scientific measure of climate
- (C) All three, because anything to do with cold and snow counts as climate
- (D) Only (ii), because it states the long-term average pattern repeated across many years

2. A pupil argues: 'Two cities both averaged exactly 600 mm of rain last year, so they must have the same climate.' What is the strongest flaw in this reasoning?

- (A) There is no flaw, since equal yearly rainfall guarantees identical climate
- (B) The flaw is that rainfall has nothing whatever to do with climate
- (C) The flaw is only that 600 mm is too small a figure to compare cities
- (D) One year's total cannot define climate, which is judged over many years, and rainfall alone ignores temperature and seasonal spread

3. Which single sentence below uses BOTH 'weather' and 'climate' correctly?

- (A) The weather of the Thar is permanently hot and dry over thousands of years, but its climate changed this afternoon.
- (B) Although today's weather in Leh is sunny, its climate is that of a cold, dry desert.
- (C) Yesterday's climate in Mumbai was humid, while its weather over thirty years is mild and even.
- (D) A place's weather never changes, whereas its climate shifts from hour to hour.

4. A scientist has only one week of readings for an unfamiliar town and is asked for its climate. The most honest answer is that she:

- (A) can state the climate exactly, since one careful week is enough to average
- (B) cannot yet state the climate, because climate needs records spanning many years, not one week
- (C) should describe the climate as 'changing fast' because the readings vary day to day
- (D) should call the week's pattern the town's climate and ignore longer records

5. Why is it possible for two places to share identical weather on one particular day yet have completely different climates?

- (A) It is not possible, because places with the same weather always share the same climate
- (B) Because weather is the average of many years while climate is just one day
- (C) Weather is a single day's condition while climate is the long-term average, so one matching day says nothing about their differing year-round patterns
- (D) Because climate depends only on which day of the week it is

6. Quito (almost on the Equator) and Madrid (well to the north) are compared. Quito's temperature barely changes between June and December, while Madrid swings from hot to cold. The factor most directly responsible for Madrid's larger seasonal swing is its:

- (A) lower altitude, since lower places always swing more between seasons
- (B) higher latitude, where the angle of the Sun's rays changes greatly between summer and winter
- (C) much greater distance from the Sun in winter than in summer
- (D) lack of any atmosphere to even out its temperatures

7. Why does a place exactly on the Equator receive more concentrated solar heating than a place near the pole, even though both are essentially the same distance from the Sun?

- (A) The Equator is millions of kilometres nearer the Sun, so it gets stronger rays
- (B) At the Equator the rays strike almost vertically, so a given bundle of sunlight is focused on a small patch, while near the pole the slanting rays spread the same energy over a wider area
- (C) The polar regions reflect every ray of sunlight, so none of it can heat them
- (D) The Sun shines for far fewer total hours per year on the Equator

8. Two students debate why the tropics stay warm all year. The one giving the BEST scientific reason says it is because:

- (A) the tropics are tilted permanently toward the Sun all year
- (B) the tropics are the only part of Earth with no winter clouds
- (C) the Earth's axis points straight at the tropics at all times
- (D) the Sun stays high in the sky throughout the year there, so its rays strike steeply and concentrate heat in every season

9. Singapore and a high Andean city lie at almost the same low latitude, yet the Andean city is chilly while Singapore is hot. Holding latitude equal, the factor that overrides it here is:

- (A) altitude, because temperature falls steadily as one climbs to great heights
- (B) latitude, which must still differ despite the figures given
- (C) distance from the Sun, the Andean city being farther out
- (D) the Andean city receiving slanting rays because it is high up

10. Near the poles the Sun's rays must travel a longer path through the atmosphere than at the Equator. This longer path matters because along it the sunlight:

- (A) is weakened as more of its energy is scattered and absorbed by the thicker layer of air it crosses
- (B) speeds up and arrives carrying extra heat
- (C) is bent downward so it strikes the ground vertically after all
- (D) gains heat from the cold polar air it passes through

11. Two towns sit at the same latitude: Town P at 2,500 m on a plateau, Town Q at 300 m in a nearby plain. A pupil predicts Town P is WARMER 'because it is higher and so nearer the Sun.' The best correction is that Town P is actually:

- (A) warmer, because the pupil is right that height brings you nearer the Sun
- (B) the same temperature, because they share a latitude
- (C) cooler, because temperature falls with altitude as the air thins and lies farther from the ground that warms it
- (D) cooler, but only because higher places are always nearer the poles

12. On a single tall mountain the foot is warm and tropical while the summit is frozen, all within a few kilometres horizontally. This vertical change of climate is caused chiefly by:

- (A) latitude, which changes sharply over those few kilometres
- (B) altitude, since the air cools steadily with height regardless of the unchanged latitude
- (C) distance from the sea increasing rapidly up the slope
- (D) the summit being much closer to the Sun than the foot

13. The general rule that air grows colder with height best explains which of these observations?

- (A) Coastal cities have milder summers than inland cities at the same latitude
- (B) Deserts behind tall mountains receive very little rain
- (C) Snow-capped peaks tower over green, warm valleys that lie at the same latitude just below them
- (D) Equatorial regions stay warm in every season

14. If you were to climb steadily from a hot coastal plain to a high mountain pass on the same day, the expected trend in air temperature is that it would:

- (A) fall fairly steadily as you gain height
- (B) rise steadily because you move closer to the Sun
- (C) stay exactly constant because your latitude does not change
- (D) rise at first and then suddenly freeze at the very top for no clear reason

15. Two cities lie at the same latitude and the same low altitude. City J is coastal; City K is deep in the continental interior. Which prediction about their yearly temperature range is most sound?

- (A) City J will have the wider range, because the sea makes coasts more extreme
- (B) Both will be identical, since latitude and altitude already match
- (C) City K will have a smaller range, because interiors are always mild
- (D) City K will have the wider range, because away from the sea land heats and cools sharply

16. Water moderates the climate of coastal places mainly because, compared with land, water:

- (A) warms up and cools down far faster, dragging the coast to extremes
- (B) reflects nearly all sunlight, so the coast receives no heat
- (C) warms up and cools down much more slowly, so it never reaches the extremes that land does
- (D) holds a fixed temperature that never varies at all

17. A place with an unusually SMALL gap between its hottest and coldest months, all else being typical, most likely owes that small range to being:

- (A) deep in a dry continental interior
- (B) on a very high, exposed plateau far from any coast
- (C) close to the moderating sea, which evens out the temperature
- (D) at a very high latitude with strong seasonal sun-angle changes

18. Two cities share the same latitude and altitude, but City M (inland) shows a 22 degree annual range while City N (coastal) shows only 9 degrees. The clearest single explanation is that:

- (A) City N is far nearer the Equator than City M
- (B) City M sits at a much greater altitude than City N

- (C) City N receives no summer sunshine, keeping its range low
- (D) the sea moderates City N, while City M, far inland, swings widely through continentality

19. Why does continentality (distance from the sea) tend to produce BOTH hotter summers AND colder winters in a single interior place?

- (A) Interiors are nearer the Sun in summer and farther in winter
- (B) The sea actively heats the interior in summer and chills it in winter
- (C) Interiors switch latitude between summer and winter
- (D) Without the slow-changing sea nearby, the land heats strongly in summer and loses its heat strongly in winter, pushing both extremes apart

20. Cherrapunji and Mawsynram receive among the world's heaviest rains because the moist monsoon winds reaching them are:

- (A) descending the dry leeward slopes and warming as they fall
- (B) blowing in from the dry Thar Desert and carrying its sand
- (C) cold winter winds off the Himalayas dropping their snow as rain
- (D) funnelled against steep hills and forced to rise, cooling so their moisture condenses into torrential rain

21. The west coast of India is drenched while the Deccan plateau just inland stays comparatively dry. Reordering this as cause and effect, the correct chain is:

- (A) moist winds meet the Western Ghats, are forced up and rain on the windward slopes, then cross the crest as dry air, leaving the Deccan in a rain shadow
- (B) the Deccan lies nearer the Equator, so it heats up and repels the rain clouds
- (C) the west coast stands higher than the Ghats, so rain collects there first
- (D) the Deccan has no rivers, so no rain can form over it

22. Orographic rainfall forms on a mountain's windward slope through which correct sequence of physical changes in the rising air?

- (A) the air rises, expands and cools, can hold less water vapour, so the vapour condenses and falls as rain
- (B) the air rises, is compressed and warms, holds more vapour, and stays dry
- (C) the air rises, speeds up and instantly dries out without cooling
- (D) the air rises, then sinks back and rains on the same windward face it climbed

23. The leeward side of a range is called a 'rain shadow' for which reason?

- (A) the mountain casts a literal shadow that blocks the sunlight rain needs
- (B) the air has already shed its moisture as rain on the windward side, so it arrives dry on the lee
- (C) rain refuses to fall on land that is too shady
- (D) the leeward side faces directly into the moisture-laden wind

24. On a map of July rainfall the windward Konkan coast is soaked while the Deccan behind it is dry, and the north-eastern hills are extremely wet while the Brahmaputra valley below is less so. The single idea tying ALL these contrasts together is:

- (A) relief lifting moist monsoon winds, giving heavy rain where the air is forced up and far less rain in the rain shadow behind
- (B) latitude alone, since rainfall simply rises as you near the Equator
- (C) altitude alone, since higher land is always wetter than lower land
- (D) distance from the sea alone, since the farther inland you go the wetter it gets

25. A hiker crosses a ridge: the climb up the seaward slope is misty and dripping, but within a few kilometres the far valley is sunny and parched. The best explanation for this rapid change is:

- (A) the rain-shadow effect, as moist air rains on the windward slope and descends dry on the leeward side
- (B) a sudden large change of latitude over those few kilometres
- (C) the far valley lying much closer to the Equator
- (D) different soils on the two sides controlling the rainfall

26. If the Western Ghats were entirely flattened, the most likely consequence for the rainfall pattern would be that:

- (A) the west coast would rain even more heavily than before
- (B) the entire peninsula would turn into a frozen tundra
- (C) rainfall everywhere would stay exactly the same
- (D) the west coast would lose its heavy rain because nothing would lift and wring out the moist winds, while more moisture might reach the now less-shadowed interior

27. The hot, dry 'loo' that scorches Rajasthan in summer and the chilling winds that bring cold waves to the plains in winter together best illustrate that winds influence a region by:

- (A) physically changing the latitude of the region they reach
- (B) lifting the region to a higher altitude as they pass
- (C) moving the region nearer to or farther from the sea
- (D) carrying the temperature and moisture qualities of the lands they have travelled across

28. Why is the summer 'loo' hot and dry rather than cool and moist?

- (A) it blows in off the cool, moist Arabian Sea
- (B) it blows across hot, dry desert and continental land, taking on that heat and dryness
- (C) it descends from the snow-covered Himalayan peaks
- (D) it rises from underground hot springs beneath the plains

29. A cold wave grips the northern plains when winter winds spill down over the Himalayas. The reason these winds chill the plains is that they have:

- (A) crossed the warm Bay of Bengal and absorbed its heat
- (B) passed over cold, snowy high ground and arrive carrying that cold
- (C) blown in straight off the hot Thar Desert
- (D) risen warm from the heated soil of the plains themselves

30. Imagine a wind that has just crossed a wide warm sea before reaching the coast. Compared with a wind that has just crossed a hot dry desert, the sea wind will most likely be:

- (A) drier, because seas remove moisture from the air
- (B) colder than the desert wind in every season
- (C) identical, because the surface a wind crosses does not affect it
- (D) moister, because it picked up water vapour over the warm sea

31. The entire Indian monsoon system ultimately depends on one physical fact about land and sea. That fact is:

- (A) the Moon's gravity drags the monsoon clouds toward India each summer
- (B) land heats up and cools down faster than the ocean, so seasonal pressure differences build up between them
- (C) rivers push warm air upward and pull sea winds inland
- (D) the Equator stays at a single fixed temperature throughout the year

32. In summer the interior of Asia heats intensely. The pressure that forms there, and the wind it draws in, are:

- (A) high pressure, which pushes dry winds out toward the sea
- (B) low pressure, into which dry winds blow outward to the sea
- (C) high pressure, which draws moist sea air inland
- (D) low pressure, into which moist sea winds blow from the south-west

33. Why does the south-west monsoon bring India's main rains while the north-east monsoon mostly does not?

- (A) the south-west winds cross warm seas and arrive moisture-laden, while the north-east winds blow off the dry land toward the sea
- (B) the south-west winds come from the dry Thar while the north-east winds come from the sea
- (C) the south-west winds are colder, and cold winds always carry more rain
- (D) the south-west winds drop straight down from the upper atmosphere carrying rain

34. The sudden, dramatic onset of heavy rain after weeks of pre-monsoon heat is called the 'burst' of the monsoon. It feels so abrupt because:

- (A) the Sun suddenly stops shining over India
- (B) the rain-bearing south-west winds arrive in force and unleash steady heavy rain after the long dry build-up

- (C) the rivers suddenly rise and spill into the air as rain
- (D) the dry north-east winter monsoon arrives all at once

35. The summer monsoon usually reaches Kerala in early June and then spreads. The correct description of its advance is that it moves:

- (A) from the far north southward, reaching Kerala last in September
- (B) from the southern tip northward, covering the whole country by about mid-July
- (C) from the east coast outward into the surrounding seas
- (D) everywhere at exactly the same instant in early June

36. The summer monsoon has been likened to 'a sea breeze the size of a country.' The likeness holds because both are driven by:

- (A) the Moon's gravitational pull on ocean water
- (B) rivers warming the air directly above their channels
- (C) warm land creating low pressure that pulls in cooler, moister air from over the sea
- (D) snow melting on high mountains and releasing winds

37. In winter the north-east monsoon blows out from the land toward the sea. The pressure pattern producing this is:

- (A) low pressure over the cold land, drawing sea air inland
- (B) high pressure over the sea, pushing moist air onto the land
- (C) equal pressure everywhere, so winds blow at random
- (D) high pressure over the cold land and lower pressure over the warmer sea, so air flows seaward

38. Most of the north-east winter monsoon is dry, yet Tamil Nadu's coast gets notable rain at that time because the winds reaching it have:

- (A) crossed the Thar Desert and gathered moisture there
- (B) blown straight down the Himalayas carrying snow-melt
- (C) crossed the Bay of Bengal and picked up moisture before striking the coast
- (D) come fresh off the Arabian Sea from the west

39. The deepest reason the monsoon winds reverse direction between summer and winter is that:

- (A) the land-sea temperature and pressure pattern flips: land is low pressure in hot summer and high pressure in cold winter
- (B) the Earth tilts its axis the opposite way twice each year
- (C) the Western Ghats physically swing around with the seasons
- (D) the Moon switches the direction of its orbit each season

40. The 'retreating monsoon' season is the period when the:

- (A) south-west monsoon first bursts over Kerala
- (B) summer heat first builds up before any rain has fallen

(C) rivers retreat back up into the Himalayas

(D) south-west monsoon weakens and withdraws, bridging the rains and the dry winter

41. Place the phases of the Indian monsoon year in their correct order: hot dry summer, rainy south-west monsoon, retreating monsoon, dry winter. Starting from the dry winter, the cycle runs:

(A) dry winter, then rainy south-west monsoon, then hot dry summer, then retreating monsoon

(B) dry winter, then retreating monsoon, then rainy south-west monsoon, then hot dry summer

(C) dry winter, then hot dry summer, then rainy south-west monsoon, then retreating monsoon

(D) dry winter, then hot dry summer, then retreating monsoon, then rainy south-west monsoon

42. India is often said to have a 'fifth season' beyond the usual four. The reason this rainy season is singled out is that:

(A) India copied a five-season calendar directly from Europe

(B) India has no winter at all, leaving room for a fifth season

(C) the monsoon rains dominate Indian life so strongly that they form a distinct season of their own

(D) every Indian festival demands its own separate season

43. The six traditional ritus divide the Indian year more finely than the four-season scheme chiefly because they:

(A) track the monsoon-ruled year more closely, recognising transitions the four-fold scheme lumps together

(B) were invented only recently for the tourist trade

(C) ignore the monsoon entirely and follow the Sun alone

(D) are simply the four European seasons renamed

44. Why do the same seasons return year after year in India?

(A) the Sun grows hotter every summer and cooler every winter

(B) the Earth revolves around the Sun on its tilted axis, so the same pattern repeats every year

(C) the monsoon winds choose to reverse on fixed festival dates

(D) rivers flood at the same time each year and create the seasons

45. Why is the region beyond the high Himalayas, such as Ladakh, called a 'cold desert'?

(A) it is hot and dry like the Thar, just given a different name

(B) it is cold only because it lies very near the North Pole

(C) the mountains block the rain-bearing winds so it stays dry, and its great altitude keeps it bitterly cold

(D) it is dry because the nearby sea pulls all the moisture away

46. Why does the flat Thar suffer searing dry winds while the lands behind the Himalayas are spared the worst of the icy Central-Asian winds? Taken together, these two facts reveal the climate role of:

(A) latitude alone, since both regions lie at the same distance from the Equator

(B) distance from the sea alone, with no role for any landform

(C) relief (topography), since flat openness exposes the Thar while the mountain wall shields the lands behind it

(D) the position of the Sun alone, regardless of the land's shape

47. Match each region to the factor that most powerfully shapes its climate: (i) Ladakh's cold desert, (ii) Mumbai's mild range, (iii) Cherrapunji's deluge, (iv) Kanniyakumari's year-round warmth. The best matching is:

(A) (i) low latitude, (ii) high altitude, (iii) distance from the sea, (iv) relief

(B) (i) nearness to the sea, (ii) latitude, (iii) altitude alone, (iv) winds only

(C) (i) high altitude plus rain shadow, (ii) nearness to the moderating sea, (iii) relief lifting moist winds, (iv) low (near-equatorial) latitude

(D) (i) winds only, (ii) rain shadow, (iii) low latitude, (iv) altitude

48. Srinagar (far north, in the mountains) is far colder than Kanniyakumari (far south, by the sea). Which two factors combine MOST powerfully to make Srinagar so cold?

(A) its low latitude and its nearness to the sea

(B) its high latitude (weaker, slanting sun) and its great altitude both push the temperature down

(C) its position right on the Equator with constant overhead Sun

(D) the moist sea winds that blow steadily over it all year

49. 'Monsoon failure' means the seasonal rains are weak or insufficient. The most likely chain of consequences for India is:

(A) rivers overflow, widespread floods spread, and harvests boom

(B) crops wither, water grows scarce, harvests shrink, and food prices rise

(C) winters turn much warmer, glaciers grow, and water becomes plentiful

(D) the six ritus vanish and the seasonal calendar collapses

50. Why does a weak monsoon hit India's economy especially hard?

(A) a large share of farming depends on monsoon rain, so poor rains cut harvests, reduce rural incomes and raise food prices

(B) festivals are cancelled by government order during dry years

(C) the sea stops moderating the coastal cities when the rains fail

(D) the Himalayas block trade routes whenever it does not rain

51. Trace the chain from a distant Pacific cause to a local Indian effect: 'warm Pacific water shifts eastward, rearranging tropical winds, leading to ____.' The most likely next link is:

- (A) record floods across India that very year
- (B) the Himalayas growing several metres taller
- (C) a weakened monsoon, so India receives less rain and crops and water supplies are stressed
- (D) the sea ceasing to moderate India's coastline

52. Every few years the monsoon is disrupted by a periodic warming of the central and eastern Pacific Ocean that rearranges tropical winds. This recurring event is known as:

- (A) El Nino
- (B) the loo
- (C) a rain shadow
- (D) the burst of the monsoon

53. A cyclone forms over warm seas when the air pressure at its centre becomes:

- (A) much higher than its surroundings, pushing air away from the centre
- (B) much lower than its surroundings, so air and moisture rush in and spiral upward
- (C) exactly equal to its surroundings everywhere
- (D) colder than the land in the middle of winter

54. During a cyclone the wind suddenly drops and the sky clears overhead. The safest interpretation is that:

- (A) the storm has fully ended and it is now safe to go outside
- (B) the cyclone has turned into a harmless breeze
- (C) the calm eye is passing over, and fierce winds will soon return from the opposite direction
- (D) the calm eye is the most destructive part and one must flee through it

55. Where in a cyclone are the most violent winds found?

- (A) in the calm, clear eye at the very centre
- (B) hundreds of kilometres out beyond the storm
- (C) only after the storm has completely passed
- (D) in the dense ring of cloud, the eyewall, surrounding the calm centre

56. Why does the calm 'eye' of a cyclone trick people into thinking the danger has passed?

- (A) because the eye is in fact the part with the strongest winds, which feel calm
- (B) because once the eye arrives the cyclone can never strengthen again
- (C) because the eye marks the point where the cyclone has left the sea for good
- (D) because the eye brings a sudden lull with clear sky, but the fierce eyewall winds return once it moves on

57. Burning fossil fuels is said to thicken Earth's natural 'greenhouse blanket.' The correct cause and effect is that the extra greenhouse gases:

- (A) trap more outgoing heat, so the planet warms
- (B) let more heat escape, so the planet cools
- (C) block incoming sunlight, so the planet freezes
- (D) remove the atmosphere, so there is no climate at all

58. Why does the NATURAL greenhouse effect actually make life on Earth possible?

- (A) greenhouse gases trap part of the Sun's heat, keeping the planet warm enough for living things
- (B) they block all of the Sun's rays, keeping Earth cool
- (C) they cool the planet to below freezing, which life prefers
- (D) they directly create the monsoon winds that water the land

59. Place these links in order for the chain 'a factory burns coal, so a distant farmer's monsoon crop fails':

- (A) coal burnt, sea cools, monsoon strengthens, crop thrives
- (B) coal burnt, rivers freeze, no irrigation, crop fails
- (C) coal burnt, carbon dioxide released, extra heat trapped, climate shifts, monsoon disrupted, crop fails
- (D) coal burnt, forests grow, more rain, crop floods

60. Two responses to climate change are 'mitigation' and 'resilience.' Which statement correctly distinguishes them, and gives a correct example of each?

- (A) mitigation slows the warming by cutting emissions, for example expanding renewable energy; resilience helps communities withstand impacts, for example building flood defences
- (B) mitigation builds flood walls to cope with impacts; resilience cuts emissions to slow warming
- (C) both simply mean planting more trees and nothing else
- (D) mitigation ignores the cause of warming while resilience ignores its effects

Solutions — Social Science (Class 7), Chapter 3: Climates of India — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	C	21	A	31	B	41	C	51	C
2	D	12	B	22	A	32	D	42	C	52	A
3	B	13	C	23	B	33	A	43	A	53	B
4	B	14	A	24	A	34	B	44	B	54	C
5	C	15	D	25	A	35	B	45	C	55	D
6	B	16	C	26	D	36	C	46	C	56	D
7	B	17	C	27	D	37	D	47	C	57	A
8	D	18	D	28	B	38	C	48	B	58	A
9	A	19	D	29	B	39	A	49	B	59	C
10	A	20	D	30	D	40	D	50	A	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Climate is the average atmospheric pattern built up over many years, so only (ii), which speaks of what winters are reliably like every year, describes climate. Statement (i) reports one morning's snow and (iii) reports the instantaneous temperature now, both of which are short-term weather. A single snowfall is not automatically climate, an instant thermometer reading is weather not climate, and lumping all three together ignores the time-scale that separates the two ideas.

2. (D) Climate is an average over many years and is described by temperature and its seasonal spread as well as rainfall, so matching one year's rain total proves little. The 'no flaw' option treats a single year as climate, the claim that rainfall is irrelevant is wrong because rainfall is a key climate element, and the size of the figure is not the issue, the time-scale and the missing temperature data are.

3. (B) Weather is the short-term condition and climate the long-term average, so 'today's weather is sunny' and 'its climate is a cold dry desert' use both words correctly. The other

three swap the meanings: calling a thousand-year pattern 'weather' and an afternoon change 'climate', calling yesterday's condition 'climate' and the thirty-year average 'weather', and claiming weather is fixed while climate changes hourly all reverse the correct time-scales.

4. (B) Climate is the average pattern over many years, so a single week's data is far too short to establish it; the honest answer is that she cannot yet say. One week is not enough to average a climate, day-to-day variation is ordinary weather rather than a fast-changing climate, and treating one week as the climate ignores the long records that the definition requires.

5. (C) A single day's matching weather is just one snapshot, whereas climate is the long-term average, so the two places can still differ greatly across the year. The 'not possible' option wrongly equates one day with the whole pattern, the second option reverses the definitions of weather and climate, and the day of the week has nothing to do with either.

6. (B) Latitude controls how much the Sun's angle changes through the year; far from the Equator the angle swings widely between seasons, giving Madrid its hot summers and cold winters, while near-equatorial Quito stays even. Altitude is not the seasonal-swing factor, Earth's tiny change of distance from the Sun is negligible against 150 million km, and Madrid certainly has an atmosphere.

7. (B) Because Earth is so distant, the difference in distance is negligible; what matters is the angle. Vertical rays at the Equator concentrate heat on a small area, while slanting polar rays spread it thinly (and pass through more air). The 'millions of kilometres nearer' claim is the classic distance fallacy, the poles do absorb some sunlight rather than reflecting all of it, and equatorial regions actually receive more, not fewer, hours of strong overhead sun.

8. (D) Near the Equator the Sun stays nearly overhead in every season, so steep rays keep heating concentrated and temperatures high year-round. The tropics are not permanently tilted toward the Sun (Earth's tilt is fixed and the regions facing the Sun change), clouds are not what keeps them warm, and Earth's axis does not point at the tropics; the steady high sun-angle is the real reason.

9. (A) With latitude held equal, the deciding factor is altitude: the high Andean city has thin, cold air and lies far above the ground that heats the air, so it is cool despite the low latitude. Latitude is given as the same, so it cannot be the cause; both cities are effectively the same distance from the Sun; and ray-slant depends on latitude and time of day, not on simply being high up.

10. (A) Slanting polar rays cross a longer, thicker slice of atmosphere, so more of their energy is scattered and absorbed before reaching the ground, weakening the heating; this adds to the spreading effect of the slant. Light does not speed up to carry extra heat,

bending does not make slanting rays vertical, and passing through cold air would not add heat to the rays.

11. (C) Climbing thins the air, which holds heat poorly, and moves you away from the ground, the air's main heat source, so the higher Town P is cooler, not warmer. 'Higher means nearer the Sun' repeats the distance fallacy, sharing a latitude does not cancel a 2,200 m height difference, and altitude, not any change of latitude, is what cools Town P.

12. (B) Over a few horizontal kilometres the latitude scarcely changes, so the warm foot and frozen summit differ because of altitude: rising air thins and cools. Latitude does not change meaningfully over such a short distance, climbing does not move you appreciably nearer or farther from the sea, and the summit is not meaningfully nearer the Sun than the foot.

13. (C) Snow caps over warm valleys at the same latitude are explained by altitude: the peaks rise into thin, cold air. The milder coastal summers are due to the moderating sea, the dry desert behind a mountain is a rain-shadow (relief) effect, and the year-round equatorial warmth is a latitude effect, so only the peaks-over-valleys case is an altitude story.

14. (A) Gaining altitude thins the air and carries you away from the ground that warms it, so the temperature falls fairly steadily as you climb. It does not rise (height does not meaningfully near the Sun), it does not stay constant just because latitude is fixed, and the cooling is a steady trend with height, not a flat rise followed by a sudden inexplicable freeze.

15. (D) Away from the moderating sea, land heats and cools rapidly, so interior City K has a wider range, while coastal City J is smoothed by the slow-changing sea. The sea makes coasts milder, not more extreme; matching latitude and altitude still leaves the sea-distance difference; and interiors are known for large ranges, not mild ones.

16. (C) Because water changes temperature slowly, the sea stays cooler than the land in summer and warmer in winter, smoothing the temperatures of nearby coasts. Water heats and cools more slowly, not faster, it does not reflect away nearly all sunlight, and its temperature does vary, just gently, so 'never varies at all' is also wrong.

17. (C) A small annual range points to the smoothing influence of a nearby sea, which warms and cools slowly. A dry interior and a far-inland high plateau both give large ranges through continentality, and a high latitude gives a strong seasonal swing, so all three of those would widen, not narrow, the range.

18. (D) With latitude and altitude matched, the difference must come from distance from the sea: coastal City N is moderated, inland City M swings widely. They share a latitude so neither is nearer the Equator, they share an altitude so height is not the cause, and a

coastal city certainly receives summer sunshine, so the sea's moderation is the explanation.

19. (D) An interior lacks the sea's smoothing, so land heats fully in summer and loses heat fully in winter, driving the hot and cold extremes apart and widening the range. Interiors do not change their distance to the Sun seasonally, a distant sea cannot heat or chill an interior it does not reach, and a place cannot switch its latitude between seasons.

20. (D) These north-eastern stations sit where moist summer winds are funnelled and forced sharply upward; the rising air cools, its vapour condenses, and enormous orographic rain falls. Descending leeward air warms and dries (it does not rain), the dry Thar cannot supply heavy rain, and these are warm moist summer winds, not cold Himalayan winter winds.

21. (A) The Ghats lift the moist winds, which rain on the windward seaward slopes; the air crosses the crest already dry, so the leeward Deccan lies in a rain shadow. The Deccan is not nearer the Equator, the coast is not higher than the Ghats, and rivers do not generate rainfall, so relief lifting the winds is the true cause.

22. (A) Rising air expands and cools; cooler air holds less vapour, so the excess condenses into rain on the windward slope. Rising air expands and cools rather than being compressed and warmed, it does not dry without cooling, and air that has risen passes over the crest and descends on the far side rather than raining back on the windward face.

23. (B) By the time air crosses the crest it has rained out its moisture on the windward slope, so the leeward side lies in a 'shadow' of rain and stays dry. It has nothing to do with literal sunlight shadows or shadiness, and the leeward side faces away from the moist wind, not into it.

24. (A) Every contrast pairs a wet windward upslope with a drier leeward area, the signature of relief lifting moist winds. Rainfall does not simply track latitude, height alone does not guarantee rain (it depends on windward exposure), and inland areas are generally drier, not wetter, so relief plus wind direction is the unifying explanation.

25. (A) Air forced up the windward slope cools and rains there, then sinks dry on the leeward side, so a short crossing takes you from wet to dry, the rain shadow. A few kilometres cannot change latitude or nearness to the Equator meaningfully, and soil type does not control rainfall, so relief and wind direction are the cause.

26. (D) The west coast's deluge comes from the Ghats forcing moist winds up to rain; remove the barrier and the coast loses that lift while the interior is no longer so sheltered. The coast would not rain more (the lifting is what produces the rain), the peninsula would not freeze, and removing a major rain barrier could not leave rainfall everywhere unchanged.

27. (D) The loo arrives hot and dry from desert lands while winter winds arrive cold from snowy heights, showing that winds bring the qualities of wherever they have been. Winds cannot change a region's latitude, raise its altitude, or relocate it relative to the sea; they simply transport the heat or cold and moisture they picked up en route.

28. (B) The loo crosses scorching, arid land and so arrives blistering and parched. Had it come off the sea it would be moist, coming off the Himalayas it would be cold, and it is an ordinary surface wind rather than air heated by underground springs, so its desert path is what makes it hot and dry.

29. (B) Winds carry the character of the land they cross, so air spilling over cold, snowy mountains reaches the plains cold and triggers a cold wave. Air off the warm Bay would be warm and moist, air off the hot Thar would be hot, and warm air rising from the plains would not produce a cold wave, so the snowy mountain path explains the chill.

30. (D) A wind crossing a warm sea evaporates water into itself and arrives moist, while a wind off a hot dry desert arrives parched. Seas add moisture rather than removing it, the sea wind is not necessarily colder in every season, and the surface a wind crosses very much affects its temperature and moisture, which is the whole principle being tested.

31. (B) The monsoon arises because land changes temperature faster than the sea, creating the pressure differences that drive the seasonal winds. The Moon's gravity governs tides, not monsoon winds; rivers do not power the system; and a fixed equatorial temperature has nothing to do with the seasonal land-sea pressure flip.

32. (D) Hot summer land forms low pressure, so cooler, higher-pressure moist sea air flows in from the south-west, giving the rain-bearing summer monsoon. High pressure would push air out (the winter case), low pressure cannot push air outward to the sea, and high pressure does not draw air inward, so only low pressure pulling in moist sea winds is correct.

33. (A) Summer south-west winds soak up moisture over warm oceans and rain it out on land, whereas winter north-east winds blow off dry land and stay dry. The first wrong option reverses the source of each wind, the second wrongly ties rain to coldness, and the third imagines rain falling vertically from space rather than being carried in moist sea winds.

34. (B) The burst marks the arrival of the moist south-west winds, switching a parched, scorching landscape to days of downpour quite abruptly. The Sun does not stop shining, rivers do not evaporate upward into rain, and the burst is a summer south-west event, not the dry north-east winter monsoon.

35. (B) The south-west monsoon enters at the southern tip in early June and progresses northward, blanketing the country by roughly mid-July. It does not march from north to

south, it does not move from the land out into the seas, and it does not arrive everywhere simultaneously because it takes weeks to cross India's length.

36. (C) A sea breeze forms when sun-warmed land makes low pressure and draws in cooler sea air, exactly the land-sea heating contrast that powers the monsoon on a country-sized scale. The Moon's pull governs tides, rivers do not drive these winds, and melting snow is unrelated to the land-sea pressure difference that creates both the breeze and the monsoon.

37. (D) In winter the land cools faster than the sea, becoming high pressure, while the relatively warmer sea is lower pressure, so air flows from land to sea, the dry north-east monsoon. The other options put low pressure or no pressure difference over the cold land or high pressure over the sea, none of which would produce the observed offshore winter winds.

38. (C) Offshore winter winds are mostly dry, but those that travel over the Bay of Bengal absorb moisture and shed it on the south-east coast. A desert supplies dryness not moisture, the Himalayas lie to the north and not in this wind's path to the south-east coast, and the Arabian Sea is on the west, not the source of the north-east flow reaching Tamil Nadu.

39. (A) In summer hot land is low pressure and pulls sea air in; in winter cold land is high pressure and pushes air out, and this seasonal flip reverses the winds. Earth's tilt is fixed (it does not flip), mountains do not swing around, and the Moon's orbit is irrelevant to the monsoon, so the land-sea pressure flip is the real cause.

40. (D) The retreating (autumn) season is when the south-west monsoon weakens and pulls back, linking the rainy season to the dry winter. It is not the burst (the arrival), it follows rather than precedes the rains so it is not the pre-monsoon heat build-up, and it has nothing to do with rivers moving.

41. (C) The year runs from cool dry winter, into the hot dry pre-monsoon summer, into the rainy south-west monsoon when the burst arrives, and finally the retreating monsoon as the rains withdraw before winter returns. The other orders break this cycle by placing the rains before summer, the retreat before the rains, or the retreat before the rains arrive at all.

42. (C) The monsoon so thoroughly governs Indian weather, farming and life that the rainy season stands out as a distinct season, the familiar 'fifth season.' It was not borrowed from Europe, India does have a winter, and festivals follow the seasons rather than each creating its own.

43. (A) The six-ritu scheme reads the monsoon-dominated Indian year in finer detail, recognising seasons the four-season model misses. It is ancient rather than recent, it is

built around the monsoon rather than ignoring it, and six categories cannot be a mere renaming of four.

44. (B) Seasons recur because Earth completes one revolution around the Sun each year while tilted on its axis, repeating the pattern annually. The Sun's output does not rise and fall seasonally, festival dates follow the seasons rather than commanding the winds, and river floods result from the seasons rather than causing them.

45. (C) Ladakh lies in the rain shadow of the Himalayas, so it gets little rain (a desert), and its high altitude makes it freezing (cold), hence 'cold desert.' It is not a hot desert, its cold comes from altitude rather than nearness to the pole, and far from being near the sea, its dryness comes from the mountain barrier cutting off moist winds.

46. (C) The flat, open Thar lets hot dry winds sweep across it, while the Himalayan wall blocks cold northern winds from the sheltered lands, both of which are effects of relief. Latitude, sea-distance and the Sun's position do not explain why a mountain barrier shields one region while an open plain is exposed; the lay of the land is the deciding factor.

47. (C) Ladakh is cold and dry from altitude plus the Himalayan rain shadow, Mumbai's mild range comes from the moderating sea, Cherrapunji's deluge from relief forcing moist winds up, and Kanniyakumari stays warm because of its low latitude. The other matchings misassign these, for instance crediting Ladakh's cold to low latitude or Mumbai's mildness to altitude, which contradicts each region's true controlling factor.

48. (B) Srinagar is both far north, where sunlight is weaker and more slanting, and high in the mountains, where altitude keeps air cold, and these stack up to chill it. It is not at a low latitude, it is nowhere near the Equator, and it is not bathed in moderating sea winds, so the correct pair is high latitude plus altitude.

49. (B) A weak monsoon means too little rain, so crops fail, water runs short, harvests shrink and food prices climb. A failed (weak) monsoon brings drought, not floods and bumper harvests; it does not warm winters or grow glaciers; and it cannot erase the seasons, so only the drought chain follows.

50. (A) Indian agriculture relies heavily on the monsoon, so when rains fail, crops fail, food grows scarce and dear, and rural incomes fall, straining the whole economy. Festivals are not legally cancelled, the sea's moderating role does not switch off, and the Himalayas do not block trade because of a dry year, so farming's dependence on the rains is the reason.

51. (C) When the periodic Pacific warming (El Nino) shifts the winds, the monsoon tends to weaken, so India gets less rain and crops and water are stressed. A weakened monsoon brings shortfall, not record floods; it does not change mountain heights; and it has no bearing on whether the sea moderates the coast.

52. (A) El Nino is the periodic warming of Pacific waters that can shift winds and weaken the Indian monsoon. The loo is a hot summer land wind, a rain shadow is the dry leeward side of mountains, and the burst is the sudden onset of monsoon rain, none of which is the Pacific warming described.

53. (B) A cyclone is built around an intense low-pressure centre into which surrounding air and moisture spiral, feeding the storm. High pressure would push air out (the opposite of a cyclone), equal pressure produces no winds at all, and cyclones grow over warm seas rather than from winter cold over land.

54. (C) The calm, clear eye at a cyclone's centre is deceptive; once it passes, the violent winds of the far eyewall strike from the opposite side, so people must stay sheltered. The lull does not mean the storm is over or harmless, and although the eye feels safe, the dangerous part is the surrounding eyewall, not the eye, so fleeing through it is unsafe.

55. (D) The fiercest winds whirl in the eyewall, the thick cloud band ringing the calm eye. The eye itself is deceptively still, the violent winds lie within the storm rather than far outside it, and they certainly do not appear after the cyclone has gone.

56. (D) The eye's lull and clear sky make people believe the storm is over, but the violent eyewall sweeps back in once the eye passes, so the calm is dangerously deceptive. The eye is calm rather than the strongest-wind region, the cyclone does not lose its power simply because the eye has arrived, and the eye is not a sign the storm has left the sea for good.

57. (A) Greenhouse gases trap heat; adding more thickens the blanket and traps even more, warming the Earth. They do not let heat escape (that would cool it), they do not mainly block incoming sunlight, and they do not remove the atmosphere, so added greenhouse gas means more trapped heat and warming.

58. (A) Without the natural greenhouse effect Earth would be far too cold; greenhouse gases trap some of the Sun's heat and keep the planet comfortably warm. They do not block all sunlight or freeze the planet (the opposite of warming), and they do not directly create the monsoon, which arises from land-sea heating, so trapping part of the heat is the right reason.

59. (C) Burning coal releases carbon dioxide, which traps extra heat, warming and shifting the climate, which can disrupt the monsoon and make crops fail; that is the correct chain. The alternatives go wrong by cooling the sea, freezing rivers or growing forests, none of which follows from burning coal, which adds greenhouse gas and warms.

60. (A) Mitigation tackles the cause by reducing emissions (such as switching to renewables) and slowing warming, while resilience prepares communities to cope with impacts (such as floods), so the matched examples are correct. The first option swaps the

two definitions, the second over-narrows both to tree-planting, and the third wrongly claims each ignores half the problem.